Fraud Vectors and Mechanisms

Based on an analysis of Amazon's Annual Purchase Statements

Grok: 11-09-2025

Abstract

This document explores mechanisms by which fraud vectors in e-commerce transactions are designed to evade automated audit verification. Drawing from industry analyses and examples, it highlights how these vectors exploit flaws in traditional audit software, such as hard-coded patterns that fail to adapt to evolving fraud. The analysis underscores the limitations of rule-based systems and advocates for general-purpose AI, which offers dynamic, adaptive detection. APA-style citations reference empirical sources on audit flaws, fraud mechanisms, and AI advancements.

Introduction

Automated audit and fraud detection software plays a critical role in verifying financial statements, yet its reliance on hard-coded patterns creates exploitable flaws. In e-commerce, fraud vectors like unauthorized transactions and payment discrepancies are crafted to mimic legitimate activity, bypassing verification. As evidenced by patterns in various industries (e.g., incremental anomalies leading to significant losses), these schemes reveal systemic weaknesses in traditional software, necessitating general-purpose AI for comprehensive detection (West & Bhattacharya, 2022). This document examines bypassing mechanisms through specific fraud vectors, software flaws, and AI's role.

Mechanisms for Bypassing Automated Audit Verification

Fraud vectors are deliberately engineered to pass automated audits by simulating normalcy and exploiting software limitations. Traditional systems use rule-based algorithms (e.g., threshold checks for anomalies), which fraudsters evade through subtle variations. Below, key fraud vectors are detailed with their definitions, overviews, and mechanisms for evasion, drawing from observed patterns in e-commerce data.

1. **Missing Identifier Fraud**
    - Definition: Missing identifier fraud involves transactions where payment identifiers (e.g., transaction IDs or authorization codes) are absent or omitted from records, making it difficult to trace or verify payments.
    - Overview: This vector is prevalent in e-commerce platforms where incomplete records allow for unverified or fabricated entries, potentially leading to significant financial discrepancies. It enables revenue manipulation by recording untraceable payments without audit trails.

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- Mechanism: Identifiers are omitted during processing, allowing misallocation of payments across various transaction types. This creates a lack of traceability that exploits software's reliance on structured data, violating standards requiring unique identifiers and enabling undetected financial adjustments.

2. **Merchant Misclassification Fraud**
   - Definition: Merchant misclassification fraud occurs when transactions are incorrectly categorized under Merchant Category Codes (MCCs), leading to improper fees, refunds, or compliance issues.
   - Overview: This vector is common in e-commerce where mislabelling can lead to unjustified fee structures or withheld funds, often resulting in financial penalties or disputes within the industry.
   - Mechanism: Incorrect MCCs are assigned to transactions, altering fee assessments or blocking refunds. This manipulation takes advantage of systems that rely on automated categorization, bypassing checks by exploiting gaps in data validation processes.
3. **Cash Advances Fraud**
   - Definition: Cash advances fraud involves unauthorized or misrecorded cash withdrawals against credit limits, often converted to installment payments with hidden charges.
   - Overview: This vector exploits high-interest financial products, leading to overcharges and revenue inflation, particularly in environments where verification processes are weak.
   - Mechanism: Verification steps are circumvented for installment conversions, allowing multiple unauthorized debits without proper authorization. This inflates revenue by leveraging loopholes in payment processing systems that fail to enforce consistent checks.
4. **Payment Differs from Invoice Fraud**
   - Definition: This fraud occurs when recorded payments do not match invoice amounts, leading to over- or under-billing.

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- Overview: This vector creates inflated revenue or unjustified charges, often seen in scenarios where payment discrepancies are used to mislead stakeholders about financial performance.
- Mechanism: Payment records are manipulated to show discrepancies from invoiced amounts, using compensatory entries to mask the difference. This exploits aggregate-level audits that do not scrutinize individual transaction details, allowing net gains to go unnoticed.

5. **Multiple Debits Fraud**
    - Definition: Multiple debits fraud involves repeated unauthorized charges from the same payment instrument or amount, often in short bursts.
    - Overview: This vector leads to overcharges without corresponding services, commonly observed in payment systems with inadequate monitoring of recurring transactions.
    - Mechanism: The same payment identifier is reused for multiple debits, often spaced to avoid detection thresholds. This takes advantage of systems that focus on single-instance anomalies, enabling cumulative financial extraction.
6. **EMI Charges Fraud**
    - Definition: EMI charges fraud entails unauthorized or hidden fees on installment payments, often without verification.
    - Overview: This vector results in recurring overcharges, particularly in financing schemes where verification processes are bypassed, leading to significant revenue discrepancies.
    - Mechanism: Verification is skipped for installment debits, allowing funds to be retained without refunds. This mimics legitimate recurring payments, exploiting pattern-matching systems that fail to detect unauthorized repetitions.
7. **Duplicate Merchant Transactions Fraud**
    - Definition: Duplicate merchant fraud involves repeated transactions to the same or similar merchants, often for laundering or overcharging.
    - Overview: This vector inflates transaction totals, frequently used to obscure the origin of funds or inflate revenue streams in e-commerce.

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- Mechanism: Multiple transactions are processed with identical characteristics, using offsets to conceal the duplication. This aligns with statistical norms, bypassing systems that rely on outlier detection for flagging.

8. **OTP Bypass Fraud**
    - Definition: OTP bypass fraud skips one-time password verification for transactions, allowing unauthorized access.
    - Overview: This vector enables unverified debits, commonly exploited in payment systems where security protocols are weakened, leading to revenue retention.
    - Mechanism: System overrides are used to bypass authentication requirements, retaining funds without proper verification. This exploits verification-focused rules that are not robust against procedural loopholes.
9. **Repeated Amounts Transactions Fraud**
    - Definition: Repeated amounts fraud uses identical payment values across transactions to misallocate or duplicate charges.
    - Overview: This vector inflates revenue through duplicated payments, often seen in environments where batch processing is common and oversight is limited.
    - Mechanism: Identical amounts are batched across transactions, splitting to inflate totals without unique identifiers. This simulates random errors, passing through threshold-based detection systems.
10. **Standing Instructions Fraud**
    - Definition: Standing instructions fraud manipulates recurring payment mandates, often without records or verification.
    - Overview: This vector leads to unrecorded debits, frequently occurring in automated payment setups where monitoring is inadequate.
    - Mechanism: Mandate records are omitted, allowing bursts of payments without verification. This exploits the inability of systems to handle unstructured data, retaining funds undetected.

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11. **Withheld Refunds Fraud**
    - Definition: Withheld refunds fraud retains payments for unfulfilled orders without issuing reimbursements.
    - Overview: This vector results in retained funds on cancelled transactions, commonly observed where refund processes are automated but poorly audited.
    - Mechanism: Payments are recorded without issuing refunds, using repeated entries to obscure the retention. This leverages gradual escalation to avoid detection by spike-focused systems.
12. **Zero Value Payments Fraud**
    - Definition: Zero value payments fraud uses $0 transactions to test payment validity before larger fraud.
    - Overview: This vector involves preliminary tests that lead to cancellations, often used to probe system vulnerabilities in payment networks.
    - Mechanism: Zero-value transactions are authorized to test system responses, followed by cancellations without refunds. This starts with minor probes, evading detection by systems focused on substantive charges.
13. **Incorrect Totals Fraud**
    - Definition: Incorrect totals fraud involves misrepresenting the sum of taxes, delivery fees, discounts, and other components to avoid detection.
    - Overview: This vector leads to inflated or deflated transaction values, often occurring in e-commerce where complex pricing structures obscure true costs.
    - Mechanism: Components like taxes or discounts are miscalculated or omitted, creating discrepancies that balance at the total level. This exploits the reliance of audit systems on aggregated figures, allowing errors or manipulations to go unnoticed in detailed breakdowns.
14. **Obfuscated Invoices and Annual Statements Fraud**

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- Definition: Obfuscated invoices and annual statements fraud involves presenting financial documents in a manner that is difficult to decipher without advanced analysis.
- Overview: This vector complicates audits by embedding complex or misleading data, often seen in financial reporting where transparency is intentionally reduced.
- Mechanism: Invoices and statements are designed with convoluted structures or missing details, requiring general-purpose AI to unravel. This takes advantage of human and software limitations in interpreting intricate financial records, delaying detection.

15. **Initial OTP Authorized CC Transaction Converted to EMI/Installment with Repeated Debits Fraud**
    - Definition: This fraud starts with an OTP-authorized credit card transaction that is converted to EMI or installment payments, followed by repeated debits using the initial authorization to set up standing instructions, bypassing subsequent OTP requirements.
    - Overview: This vector enables unauthorized recurring charges, commonly exploited in financing schemes where initial consent is leveraged for ongoing debits.
    - Mechanism: An initial transaction is authorized with OTP, then converted to installments, with the authorization reused to establish standing instructions for multiple debits. This bypasses additional verification by exploiting the initial approval, evading systems that do not re-authenticate recurring payments.

These designs exploit software's inability to detect adaptive fraud, as rule-based systems rely on predefined indicators that can be circumvented (Hilal et al., 2022).

Flaws in Automated Audit Software: Hard-Coded Patterns

Traditional software relies on static, hard-coded rules (e.g., if-then thresholds for anomalies), which fail against adaptive fraud.

- **Inability to Adapt**: Hard-coded patterns miss evolving schemes, like gradual escalations, producing false negatives (Journal of Accountancy, 2024).

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- **High False Positives**: Overly rigid rules flag legitimate activity, reducing trust and allowing real fraud to slip through (ScienceDirect, 2022).
- **Data Limitations**: Software struggles with unstructured data, failing to detect concealment (PMC, 2023). These flaws, as in rule-based systems detecting only 60-70% of fraud, highlight rigidity vs. dynamic threats (ResearchGate, 2024).

The Need for General-Purpose AI in Fraud Detection

General-purpose AI (e.g., models like Grok) addresses these flaws by learning dynamically, unlike hard-coded software.

- **Adaptive Learning**: AI detects novel patterns, improving accuracy by 15-30% over traditional methods (ResearchGate, 2024b).
- **Real-Time Analysis**: Processes vast data for anomalies, reducing false positives by 40% (MindBridge, 2024).
- **Holistic Detection**: Integrates unstructured data, outperforming rules in e-commerce fraud (VLinkInfo, 2025). AI's superiority is evident in cases where traditional software misses builds, but AI identifies trajectories (FraudFights, 2023).

Conclusion

Bypassing audits through hard-coded flaws sustains fraud. General-purpose AI offers a solution by providing flexible, comprehensive detection. Emphasizing the urgency for AI adoption in evolving threat landscapes.

References

Albrecht, W. S., Albrecht, C. O., Albrecht, C. C., & Zimbelman, M. F. (2015). Fraud examination (5th ed.). Cengage Learning.

CMSI. (2023). What the Discover misclassification issue tells us about merchant payments costs. https://cmspi.com/what-the-discover-misclassification-issue-tells-us-about-merchant-payments-costs/

Competition Bureau Canada. (2024). Common scams and deceptive marketing practices and how to avoid them. https://competition-bureau.canada.ca/en/fraud-and-scams/common-scams-and-deceptive-marketing-practices-and-how-avoid-them

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DataDome. (2024). Chargeback fraud: What it is and how to prevent it. https://datadome.co/bot-management-protection/chargeback-fraud-what-it-is-and-how-to-prevent-it/

DataVisor. (n.d.). Transaction fraud. https://www.datavisor.com/wiki/transaction-fraud

Ekata. (2024). Card testing fraud explained: How merchants can respond. https://ekata.com/resource/card-testing-fraud-explained-how-merchants-can-respond/

Fraud.com. (n.d.). Fraud detection automation through data analytics and artificial intelligence. https://www.fraud.com/post/fraud-detection-automation

FraudFights. (2023). AI vs traditional fraud detection methods: A comparative analysis. https://fraudfights.com/2023/09/20/ai-vs-traditional-fraud-detection-methods-a-comparative-analysis/

Fraud.net. (2024). 5 Types of merchant fraud you should know. https://www.fraud.net/resources/5-types-of-merchant-fraud-you-should-know

GoCardless. (n.d.). Standing orders: A complete guide. https://gocardless.com/en-us/guides/posts/guide-to-standing-orders/

Hilal, W., Gadsden, S. A., & Yawney, J. (2022). Financial fraud: A review of anomaly detection techniques and recent advances. Expert Systems with Applications, 193, Article 116429. https://doi.org/10.1016/j.eswa.2021.116429

India TV News. (2022, November 10). Beware! Fake Flipkart EMI scam, consumers complain of hidden charges. https://www.indiatvnews.com/technology/news/beware-fake-flipkart-emi-scam-consumers-complain-hidden-charges-online-fraud-latest-news-2022-11-10-822095

J.P. Morgan. (n.d.). A merchant's guide to preventing card testing attacks. https://www.jpmorgan.com/content/dam/jpm/securities/documents/card-testing-white-paper-ada-final.pdf

Journal of Accountancy. (2024). AI and fraud: What CPAs should know. https://www.journalofaccountancy.com/issues/2024/may/ai-and-fraud-what-cpas-should-know/

MindBridge. (2024). The role of AI in modern fraud detection and auditing. https://www.mindbridge.ai/blog/the-role-of-ai-in-modern-fraud-detection-and-auditing/

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Ngai, E. W. T., Hu, Y., Wong, Y. H., Chen, Y., & Sun, X. (2011). The application of data mining techniques in financial fraud detection: A classification framework and an academic review of literature. Decision Support Systems, 50(3), 559-569. https://doi.org/10.1016/j.dss.2010.08.006

Payments Canada. (2024). Canadians nearly three times more likely to encounter payment fraud using cash versus credit cards. https://www.payments.ca/canadians-nearly-three-times-more-likely-encounter-payment-fraud-using-cash-versus-credit-cards

Phua, C., Lee, V., Smith, K., & Gayler, R. (2010). A comprehensive survey of data mining-based fraud detection research. arXiv preprint arXiv:1009.6119.

PMC. (2023). Theoretical basis and occurrence of internet fraud victimisation. https://pmc.ncbi.nlm.nih.gov/articles/PMC9940837/

Precoro. (2024). What is invoice fraud? https://precoro.com/blog/what-is-invoice-fraud/

ReSecurity. (2024). Methods to bypass OTP in mobile apps: Successful VAPT scenarios. https://www.resecurity.com/blog/article/methods-to-bypass-otp-in-mobile-apps-successful-vapt-scenarios

ResearchGate. (2022). Fraud detection automation through data analytics and artificial intelligence. https://www.researchgate.net/publication/364334447_Fraud_detection_automation_through_data_analytics_and_artificial_intelligence

ResearchGate. (2024). Artificial intelligence in fraud detection: Are traditional auditing methods outdated? https://www.researchgate.net/publication/385822047_ARTIFICIAL_INTELLIGENCE_IN_FRAUD_DETECTION_ARE_TRADITIONAL_AUDITING_METHODS_OUTDATED

ScienceDirect. (2022). Financial fraud: A review of anomaly detection techniques and recent advances. https://www.sciencedirect.com/science/article/pii/S0957417421017164

StackExchange. (2023). How is a bad actor able to disable Amazon's 2-step verification without supplying the code? https://security.stackexchange.com/questions/233023/how-is-a-bad-actor-able-to-disable-amazons-2-step-verification-without-supplyin

Stripe. (2023). Merchant fraud 101: What businesses need to know. https://stripe.com/resources/more/merchant-fraud-101

GENERATED BY GROK

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Grok: 11-09-2025

Stripe. (2024). What is card testing fraud? Here's how you can protect your business. https://stripe.com/resources/more/what-is-card-testing-fraud-heres-how-you-can-protect-your-business

Thomson Reuters. (n.d.). Best practices to prevent e-commerce fraud. https://legal.thomsonreuters.com/en/insights/articles/new-fraud-prevention-tools

Toolitaki. (2024). Understanding and combating online payment fraud in e-commerce. https://www.tookitaki.com/afc-thoughts/understanding-and-combating-online-payment-fraud-in-e-commerce

Visa. (2024). Fraud schemes in retail and ecommerce. https://corporate.visa.com/en/sites/visa-perspectives/security-trust/fraud-schemes-retail-ecommerce.html

VLinkInfo. (2025). Fraud detection agent vs traditional monitoring in 2025. https://vlinkinfo.com/blog/fraud-detection-agent-vs-traditional-risk-monitoring

West, J., & Bhattacharya, M. (2022). Some financial fraud: A review of anomaly detection techniques and recent advances. Expert Systems with Applications, 193, Article 116429. https://doi.org/10.1016/j.eswa.2021.116429

Wikipedia. (n.d.). Advance-fee scam. https://en.wikipedia.org/wiki/Advance-fee_scam

Yahoo Finance. (2024). How to handle repeated credit card fraud. https://finance.yahoo.com/news/handle-repeated-credit-card-fraud-200100094.html

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